Kuku AI Generalist Assignment - Problem 1: Find the Promos

Show: I Married a Broken Soldier. He was a Duke (50 episodes) | Prototype: CLI + Streamlit + Swappable LLM

1. Problem Choice, Framing & Explicit Assumptions

Why Problem 1: Reviewing 50 episodes by hand to find 20-60s ad hooks is a high-volume editorial bottleneck that repeats every week across languages. Because Kuku's screenplays already carry precise timestamps, speaker tags, emotions, and camera/sound cues, we can solve candidate discovery in deterministic code and reserve AI strictly for semantic judgment.

- **Single continuous windows (20-60s):** While promos can stitch multiple segments, continuous scenes preserve dramatic pacing without jarrin continuous windows aligned strictly to dialogue/action boundaries (never cutting a spoken line in half), with 2 sweet spot.
- **Up to 2 non-overlapping candidates per episode:** Gives editors 100 total candidates across the 50-episode pack, ranked globally so the acquisit test first.
- **Zero-dependency fallback:** If an API key is missing or rate-limited during a live run, every stage degrades gracefully to an offline model/heurist crashes on fresh inputs.

2. Architecture: Code vs. Self-Supervised ML vs. LLM vs. Human

Rather than dumping entire screenplays into an LLM (which hallucinates timestamps and scales poorly), we use a 4-stage funnel where cheap stages filter 16,600+ windows down to 12 finalists:

- **Stage 1 - Deterministic Parser & Window Proposer (Code, promo_finder.py):** Parses character dialogue, parenthetical emotions, ACTION, CA windows). Guarantees exact timestamps and zero split lines.
- **Stage 2a - Structural & Lexicon Rules (Code):** Scores every window on Hook (first 3s: slaps, accusations, exclamations, intense emotion tags, pe 'Yes/And'), Cliffhanger (last 6s: episode-end stings, unanswered questions, threats, penalisi (scene-start bonus, cast-size cap, early-episode bonus), and Intensity.
- **Stage 2b - Novelty 1: Self-Supervised Hook/Cliffhanger Classifier (ML, learn.py):** Every episode opening (first 3s) is a writer-chosen hook and ev positive examples per class with zero manual labelling. We tra features (blended 50/50 with rules). In 5-fold leave-episode- cliffhangers.
- **Stage 3 - Novelty 2: Cold-Viewer Simulator & Curiosity-Gap Score (LLM, judge.py):** Rules are worst at judging 'makes sense on its own.' For swappable via LLM_MODEL / LLM_BASE_URL) reads t past_confusions vs. urgent future_questions and rating stan
- **Stage 4 - Novelty 3: Position-Debiased Pairwise Tournament (LLM, rank.py):** Absolute 0-10 LLM scores are uncalibrated across episodes. The t calls). Clip A beats Clip B only if the LLM picks A when shown (Human editors make the final cut.

3. Scale & Unit Economics at Production Volume

On all 50 episodes, Stages 1-2 evaluate ~16,600 windows in 2.5 seconds on CPU ($0). Stage 3 makes 200 short LLM calls and Stage 4 makes 132 calls (~332 calls/show, ~140k input + 25k output tokens). Using Gemini Flash-Lite or Flash free/paid tiers, a full 50-episode show costs ~$0.02-$0.05 and runs in ~40 seconds with thread parallelism (and <2s on cached re-runs). At 20 new shows/week, total AI cost is under $1/week.

4. Top Ranked Promos: Which to Post First & Why

Across all 50 episodes, our pairwise tournament and composite scorer surface three distinct promo archetypes to post first on Reels/TikTok/Shorts (full 100-clip table + transcripts in outputs/promos.md):

- **#1 (Post First - Visceral Dungeon Torture): Ep 31, 02:02.00 -> 02:40.00 (38.0s, Score 6.12, 11.0/11 Wins).** Hook: '[ACTION] Catherine's smug e powerful, Catherine, you wouldn't ne branding iron inches from Juliana's ey and an unbearable visual cut.
- **#2 (Post Second - Class Humiliation & Physical Strike): Ep 24, 00:33.80 -> 01:10.50 (36.7s, Score 6.44, 8.0 Wins).** Hook: '[ACTION] The noblew drops to her knees before a punishment?' Why second: C
- **#3 (Post Third - Altar Betrayal, Series Premise): Ep 1, 01:47.50 -> 02:17.50 (30.0s, Score 6.27, 8.0 Wins).** Hook: Clarissa gloating in gold at the Cliffhanger: Groom rejects Juliana as ' third: Establishes the core series prem
- **#4 & #5 (High-Stakes Conspiracy & Scandal):** Ep 30 (00:08.80 -> 00:39.80, Score 6.00, 7.5 Wins: 'You have a rat in your cellar, Rupert... yo 01:18.50, Score 6.00, 6.0 Wins: maid Agnes catches Lady Catherine in bed with the shirtless priest cut to black).

5. Quality Checking & Where the Tool Picks Badly

All 100 output promos pass hard boundary checks (0/100 outside 20-60s, 0/100 cutting a spoken line in half). On a stratified 24-clip evaluation set (gold/gold.csv, 11 good / 13 bad), all four variants achieve Precision@6 = 1.00, with ROC-AUC improving from 0.972 (rules only) to 0.979 (rules + learned ML / full LLM) and 0.983 (full offline). Automated Sanity & Human-Labelled Benchmark (eval.py):

Inspecting the full 100-clip ranking reveals four concrete weaknesses: Honest Failure Modes (Where Our Tool Picks Badly):

- **1. Stage-direction keyword leakage on resolved scenes (Ep 16, 00:41.20 -> 01:23.19, Ranked #7):** Opens with a great hook (Alexander choking quiet, tender gaze between Alexander and J 'swelling to a dramatic cliffhanger hold', the resolution.
- **2. Quiet conversational openers rescued only by huge endings (Ep 50, 00:49.80 -> 01:18.50, Ranked #5):** Ends on arguably the best scandal since the dawn in your bedchamber Agnes's gasp.
- **3. Forced quota on low-conflict transition episodes (Ranks #85-#100):** Because --per-episode 2 forces two picks per episode, quiet world-buildi fireplace herbal chat; Ep 48 #99, Score 3.07) produce flat clips. In produc per-episode quota.
- **4. Late-series lore density (Ep 45 #12, Ep 42 #9):** Even when dramatic, late-episode reveals ('You poisoned my mother. You tortured my wife.') sp buildup.

6. AI Tools Used & Where AI Got It Wrong During Development

Tools used: Claude / Gemini in Antigravity IDE for architecture design, scikit-learn model iteration, and Gemini 3.1 Flash-Lite via REST API for Stage 3-4 runtime scoring.

- **Mistake 1 - Confounded Self-Supervised Labels:** In our first version of Novelty 1, TF-IDF + Logistic Regression achieved a suspiciously high 0 (action block) and plot words like 'ten days'. Inspecting the data showed two bugs: (a) episode ed block, so random mid-episode dialogue negatives taught the model 'ACTION = hook'; and (b romance show are tender fireplace moments (Ep 6, 7, 8), not cliffhangers! Fix: We matched nega training edges by the rule median, and shipped only the 14 show-agnostic structural features.
- **Mistake 2 - LLM List-Padding in Curiosity-Gap Prompting:** In our first test of Novelty 2 on Episode 1, Gemini returned exactly 4 past_confusions a trivia like 'Why is she wearing her mother's gown?'), which pegged our count formu prompt to restrict past_confusions strictly to plot-breaking missing context (returning [] explicit 0-10 clarity/pull ratings.